

Chapter 7.A

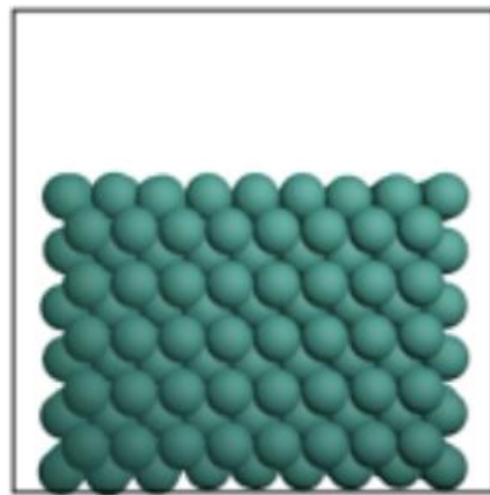
Gases

Chemical Principles I

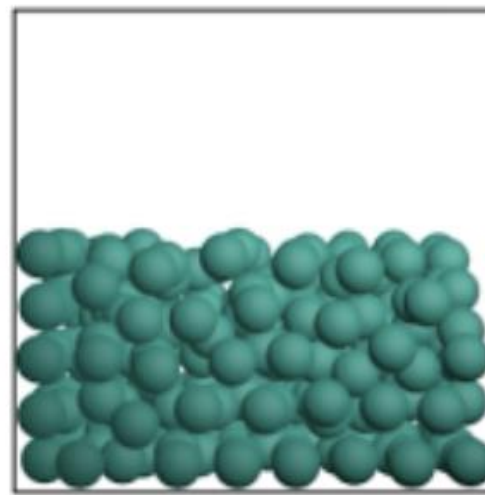
CHEM 1211K

1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using values for the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Carry out stoichiometric calculations for reactions involving gases.
5. Use the kinetic theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.
8. Discuss the circumstances that lead to deviations from ideal gas behavior.

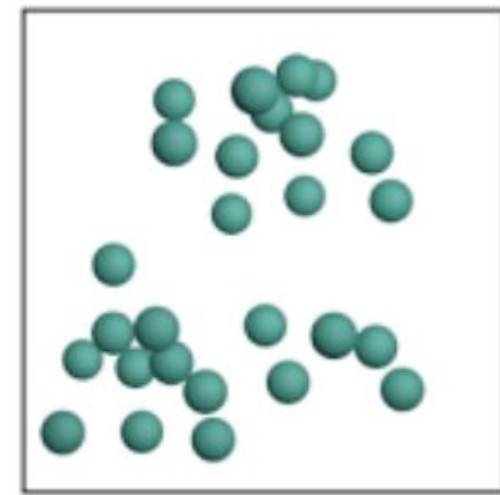
- The three common states or phases of matter are **solid**, **liquid**, and **gas**.
- Gas is the *simplest* phase of matter because particles are essentially isolated from each other.
- Gases do not have intrinsic volume; they expand to fill the volume of the container.



Solid



Liquid



Gas

Figure 7.1. Molecular-level pictures of the phases of matter.

- **Pressure** is the force exerted by a gas per unit area over which the force acts.
- **Hydrostatic pressure** is pressure exerted by a column of fluid with mass density ρ . Pressure can be expressed in “height of fluid” units.

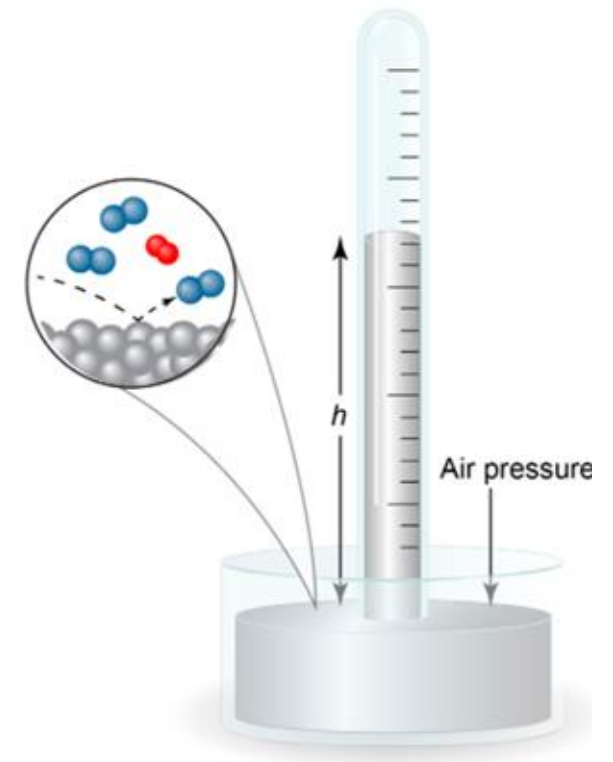


Figure 7.2. A simple barometer.

- Units of pressure include “force per area” units and “height of fluid” units
 - 1 Pascal (P)
 - 1 millimeter of mercury (mmHg)
 - 1 atmosphere (atm)
 - 1 bar
 - 1 pound per square inch (psi)

- **Standard temperature and pressure (STP)** is a specific set of temperature and pressure conditions...

7.2 Boyle's Law

- We are interested in relations between four key variables for gases: pressure (P), volume (V), number of moles (n), and temperature (T)
- **Boyle's law:** the pressure and volume of a gas are inversely proportional at constant number of moles and temperature.

$$PV = k$$

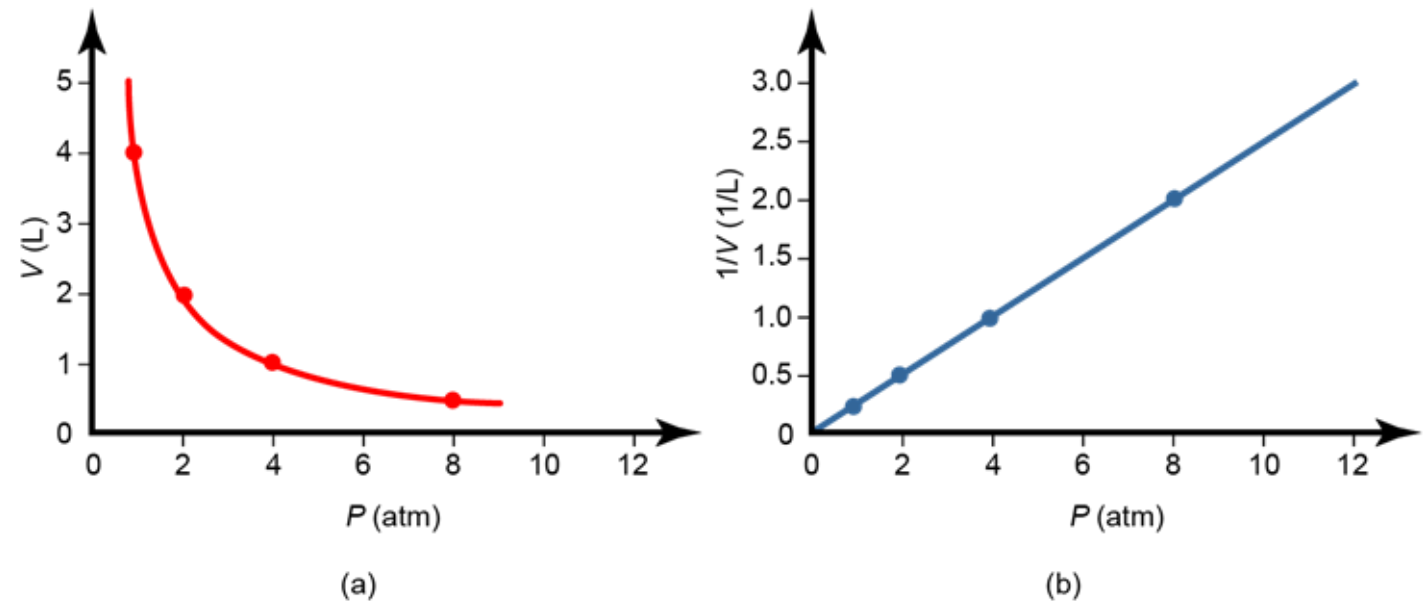


Figure 7.5. Graphs of volume versus pressure (a) and inverse volume versus pressure (b).

7.3 Charles's Law | 1 of 2

- **Charles's law:** the volume and temperature of a gas are directly proportional at constant number of moles and pressure.

$$V = kT$$

Temperature (°C)	Temperature (K)	Volume (L)
−3	270	22
−23	250	21
−53	220	18
−162	111	9

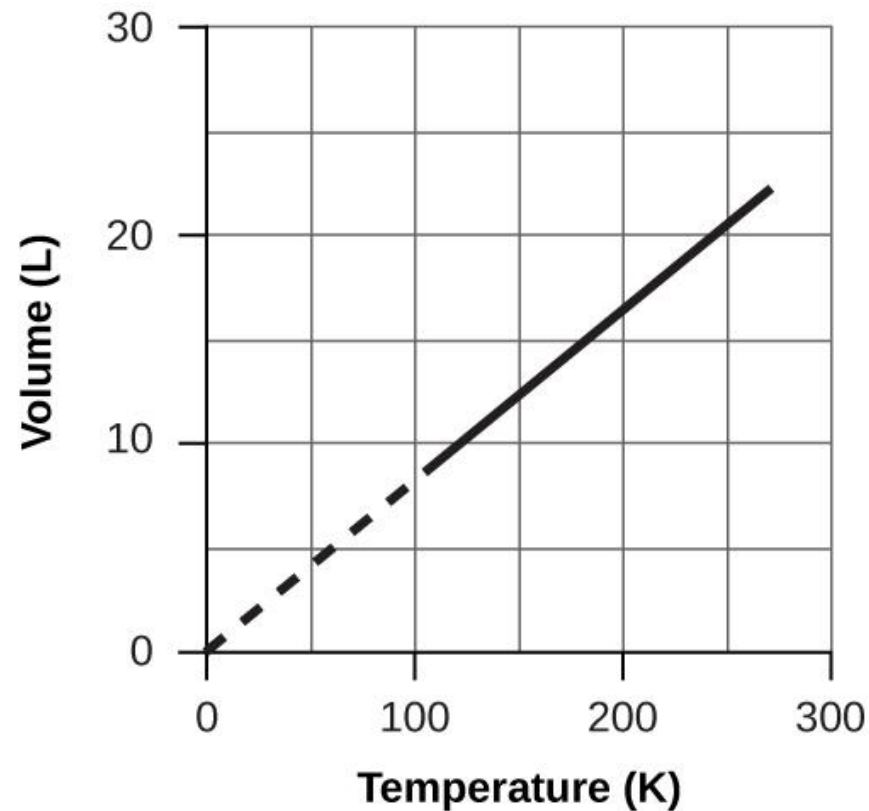


Figure 9.12, OpenStax Chemistry 2e. Volume and temperature for a gas.

- Charles's work also led to the development of the **Kelvin** temperature scale.
- Kelvin should *always* be used when working problems involving gases.

7.4 The Combined Gas Law

- The **combined gas law** incorporates the previous two laws and holds true for a gas at constant number of moles. The ratio of PV and T is constant at constant n .

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

The pressure exerted on a sample of a fixed amount of gas is doubled at constant temperature, and then the temperature of the gas in Kelvin is doubled at constant pressure. What is the final volume of the gas?

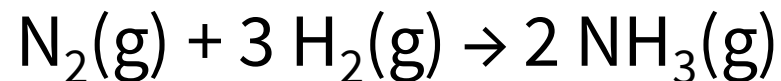
- A. Twice the initial volume
- B. Four times the initial volume
- C. Half of the initial volume
- D. One quarter of the initial volume
- E. The same as the initial volume

By what ratio does the volume of a gas increase when its temperature is increased from $9.00\text{ }^{\circ}\text{C}$ to $27.0\text{ }^{\circ}\text{C}$ at constant pressure?

- **Avogadro's law:** the volume and number of moles of a gas are directly proportional at constant pressure and temperature.

$$V = kn$$

- Avogadro's law is relevant to gas-phase reactions occurring at constant temperature and pressure (for example, inside a balloon in a thermostatic room).
- When 1.00 L N₂ is combined with 3.00 L H₂...



- The previously described **empirical gas laws** can be combined to create a universal equation of state for ideal gases known as the **ideal gas law**.

$$PV = nRT$$

- The **ideal gas constant (R)** has units of energy (!) per mole per Kelvin. The energy units can vary; the units of P and V should match the energy unit in R when we carry out calculations.

$$R = PV/nT$$

- **Partial pressure (P_i):** the pressure due to gas i in a mixture of gases.
- **Mole fraction (x_i):** moles of gas i divided by the total moles of gas in a mixture of gases.
- **Dalton's law of partial pressures**
 - The total pressure of a mixture of gases is equal to the sum of the partial pressures of the constituent gases.
- Given the total pressure of a mixture of gases and the mole fractions of the constituents, the partial pressure of each gas is the product of the total pressure and the mole fraction.

Example. Consider a gas cylinder containing 1.0 mol A, 0.4 mol B, and 0.6 mol C. If the total pressure in the cylinder is 5.0 atm, what are the partial pressures of each gas?

Container A contains a gas under 3.24 atm of pressure. Container B contains a gas under 2.82 atm of pressure. Container C contains a gas under 1.21 atm of pressure. If all these gases are put into Container D, then what is the pressure in Container D in atm?

Container A = 1.23 L

Container B = 0.93 L

Container C = 1.42 L

Container D = 1.51 L

7.9 Gases in Chemical Reactions

- The ideal gas law is a useful addition to our stoichiometry toolbox, enabling pressure or volume to “stand in” for number of moles.

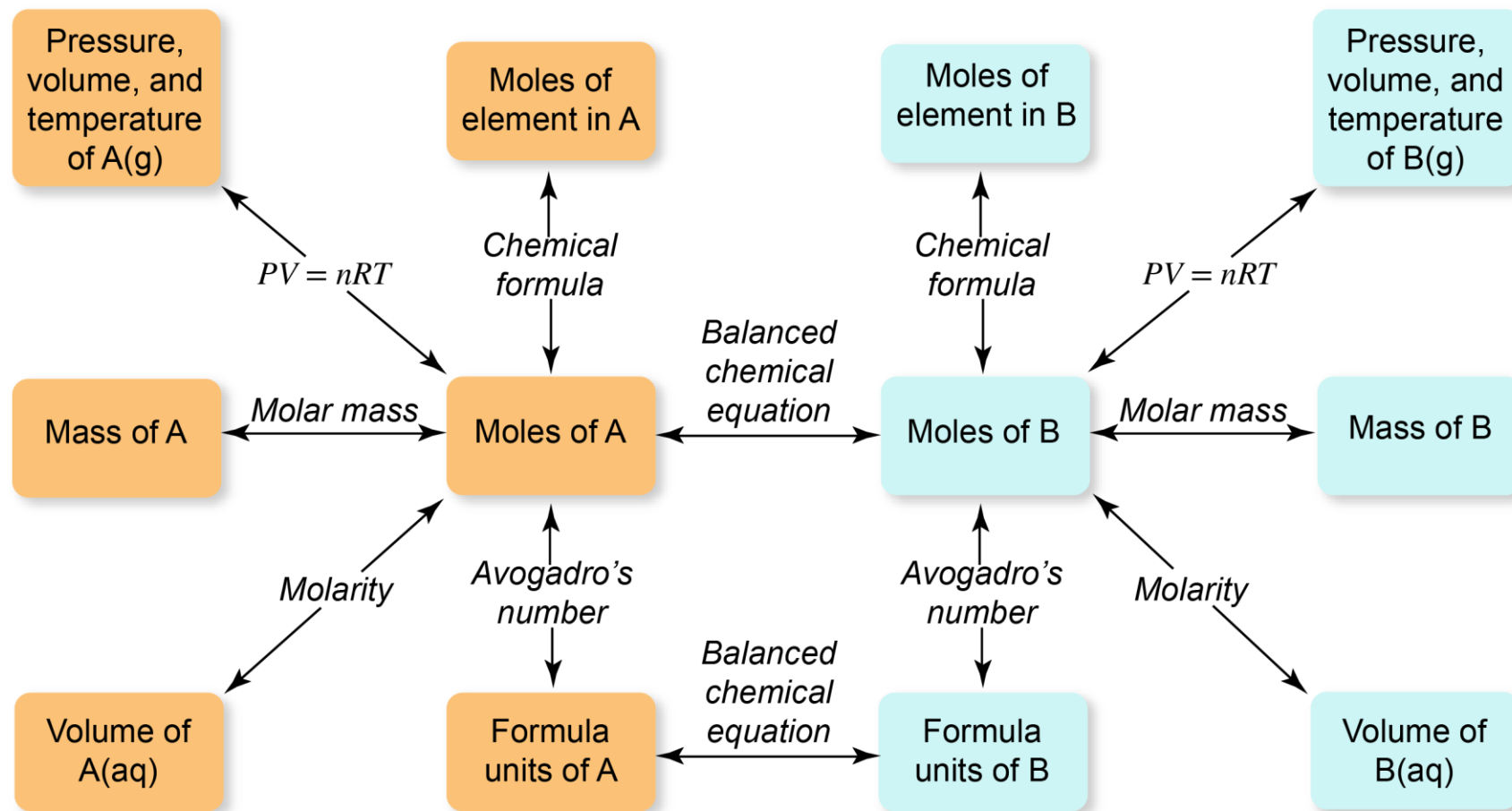


Figure 7.20. A stoichiometry toolbox for reactions involving gases.

A 5.43 g sample of an elemental diatomic gas occupies 4.38 L at 25 °C and 0.950 atm. What is the identity of the gas?
Enter the molecular formula of the gas.

Chapter 7.B

Gases

Chemical Principles I

CHEM 1211K

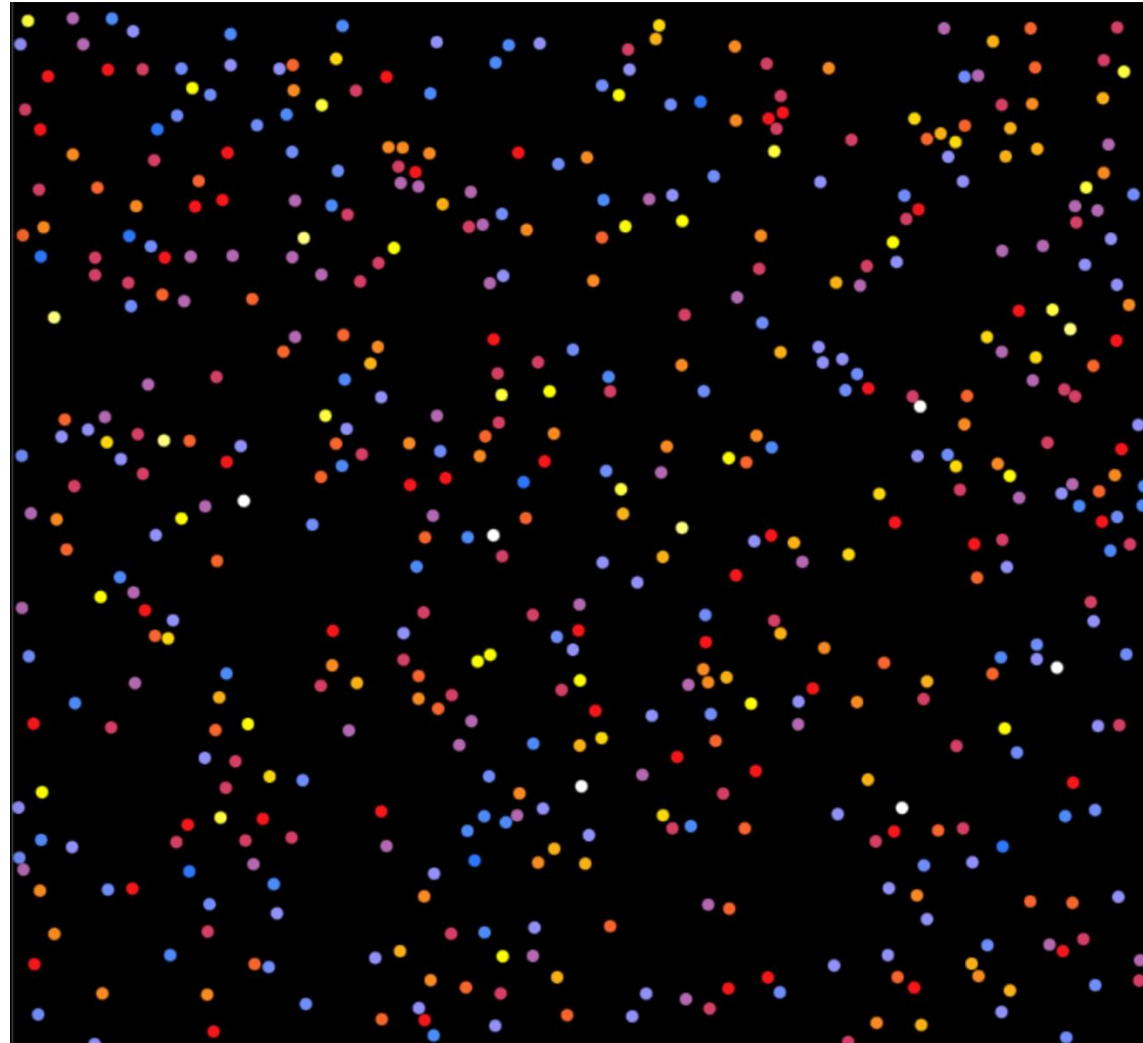
1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using the quantities of the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Predict how the pressure in a fixed-volume container will change after a complete reaction.
5. Use the kinetic theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.
8. Discuss the circumstances that lead to deviations from ideal gas behavior.

N_2O_5 is an unstable gas that decomposes according to the following balanced equation.



A sample of N_2O_5 is placed into a 5.00-L container at 298.0 K. The initial pressure inside the container is 0.400 atm. What is the $P(\text{NO}_2)$ after the sample of N_2O_5 decomposes if the temperature after reaction is 310.2 K?

- The ideal gas law follows from a model of the ideal gas based on the **kinetic theory of gases**.

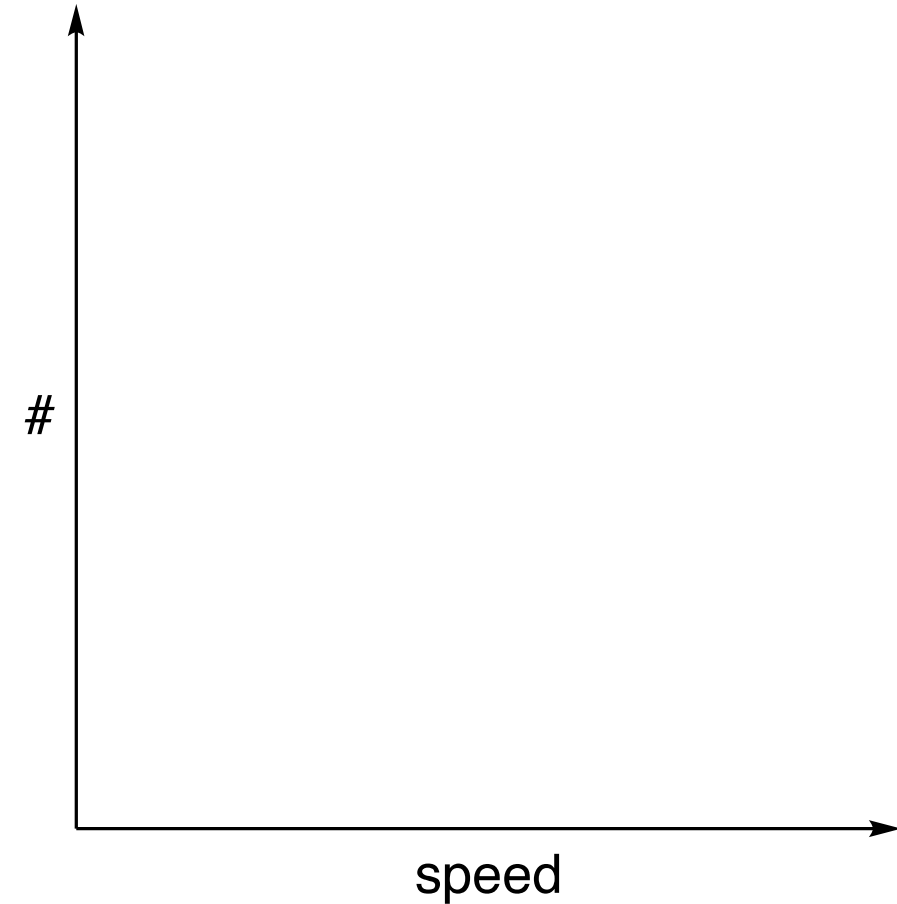


<https://falstad.com/gas>

- The kinetic theory of gases includes five key tenets or postulates.
 1. Gas particles take up no volume (i.e., particles are infinitely small points).
Every particle can access the full volume V of the container.
 2. No forces exist between particles except on collision.
Thus, all the energy of an ideal gas is kinetic energy.
 3. Collisions are perfectly elastic. No energy is dissipated or “internalized” during collisions. Collisions do not change the total kinetic energy.
 4. Particles are in constant random motion based on Newton’s laws of motion.
 5. Absolute temperature (Kelvin) is proportional to the average kinetic energy of the gas particles. This is the definition of the Kelvin scale.

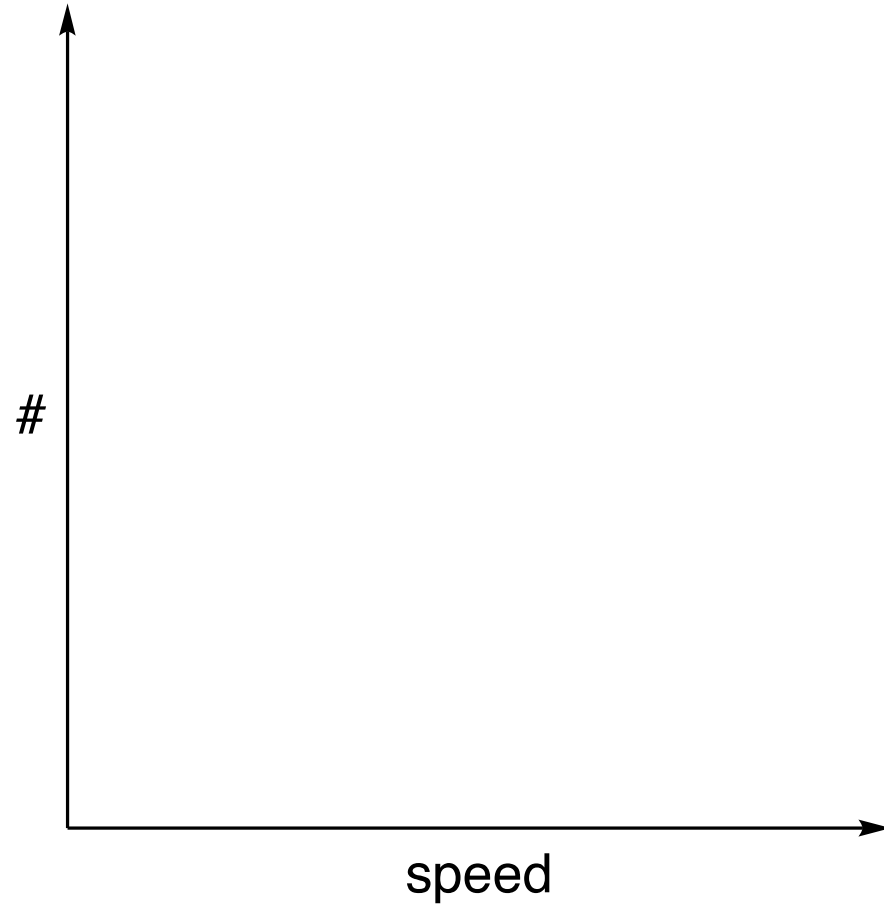
- Explaining gas pressure using the kinetic theory...
- Explaining Dalton's law using the kinetic theory...

- The distribution of particle speeds in an ideal gas is called the **Maxwell distribution**.
- The **most probable speed (v_p)** is located at the peak of the distribution.
- The **root mean square speed (v_{rms})** is the square root of the average of squared particle speeds.

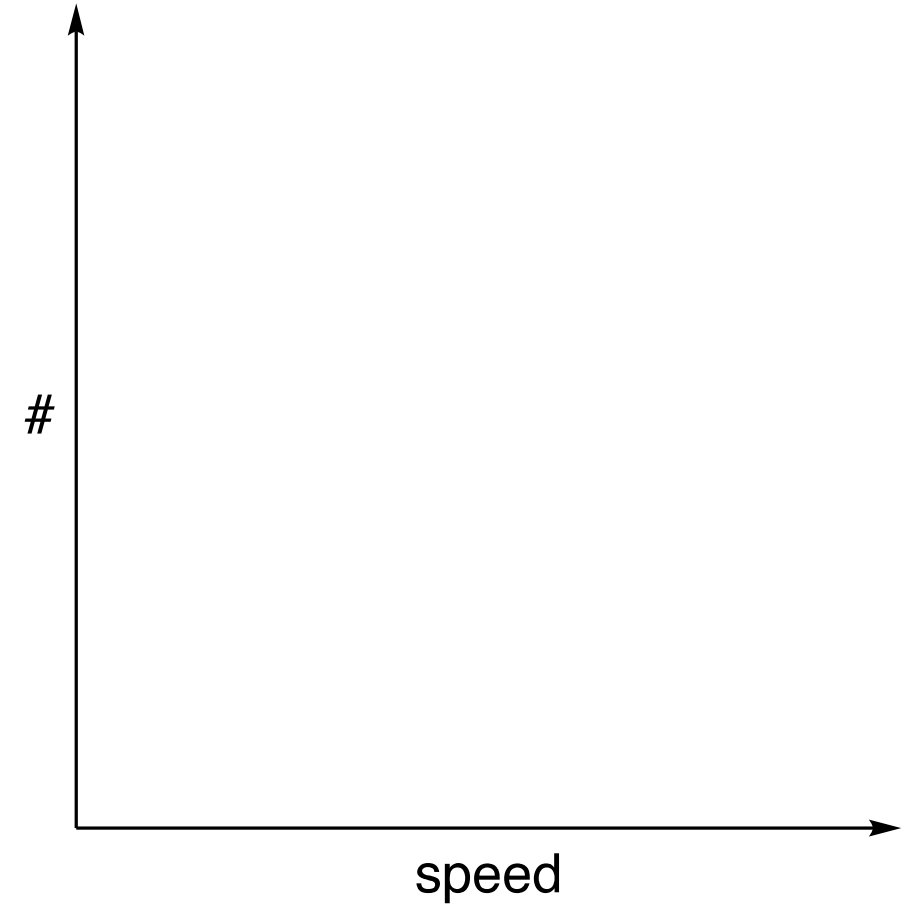


- Most probable speed and root mean square speed depend on temperature and molar mass.

Constant temperature, varying molar mass



Constant molar mass, varying temperature



- Gas particles generally travel only a very short distance before colliding with either another particle or the container wall and changing direction.
- **Mean free path** is the average distance traveled by a gas particle between collisions.

Which of the following statements is/are *true* of ideal gases?

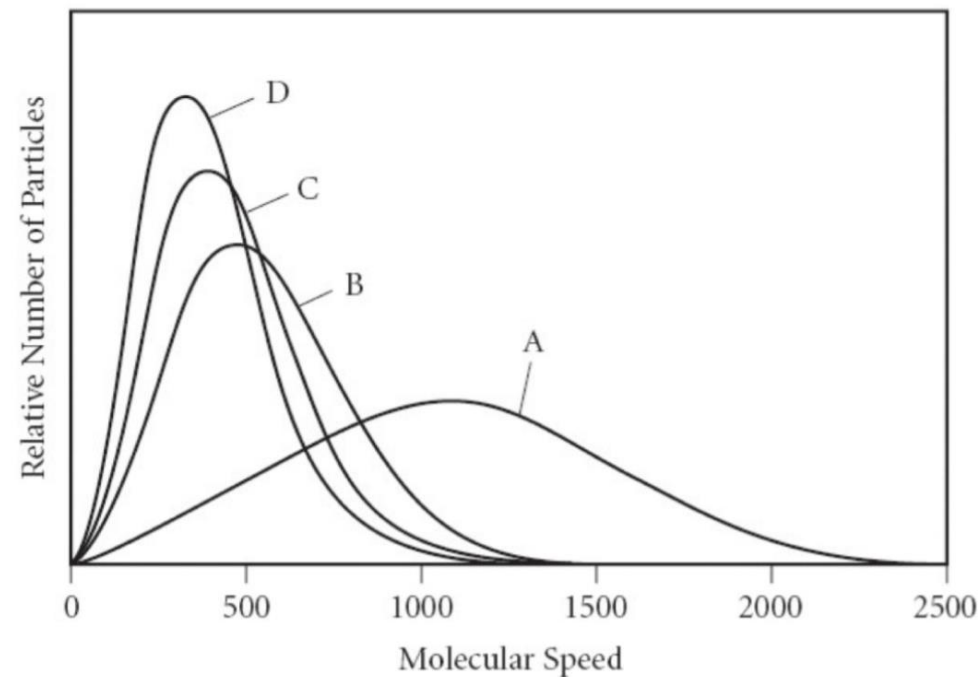
- A. When molecules of ideal gas collide, energy is exchanged between them.
- B. Molecules of F_2 gas and Cl_2 gas have the same kinetic energy at 875 K.
- C. Particles of gas are more attracted to one another at 1000 K than at 300 K because the particles at 1000 K have greater kinetic energy.
- D. Kinetic-molecular theory adequately explains why increasing the number of moles of gas in a sample increases pressure.
- E. Increasing pressure causes gases to behave more ideally because it makes the volume of gas particles appreciable with respect to the volume of the container.

- **Diffusion** is the process by which gas particles spread out in response to a gradient in concentration.
- Diffusion is faster at greater root mean square speed.

- **Effusion** is the one-way escape of a gas from a container through a tiny hole into vacuum.
- **Graham's law of effusion...**

The following graph represents four different gases at the same temperature. Which of the following statements are *true*?

- A. Molecules of the gas represented by curve A have a greater distribution of speeds due to the gas having a larger molar mass.
- B. Molecules of the gas represented by curve C have a smaller molar mass than the molecules represented by gas B.
- C. All the curves actually represent the same compound at the same temperature, but have different root mean square speeds.
- D. Molecules of the gas represented by curve D have the smallest root mean square speed and would diffuse the fastest.
- E. The molecules of gas B will effuse faster than the molecules of gas C.



● iClicker Question 7.8

A mixture of $\text{O}_2(\text{g})$ and $\text{SO}_2(\text{g})$ in a gas cylinder is allowed to effuse through a tiny orifice. Suppose that in some time frame, 1.00 mol of $\text{O}_2(\text{g})$ have effused out of the cylinder.

How many moles of $\text{SO}_2(\text{g})$ have effused out in that same time frame?

- The tenets of the kinetic theory don't always hold...
- Particularly at high pressures and low temperatures, gases tend to violate the ideal gas law (behave nonideally).

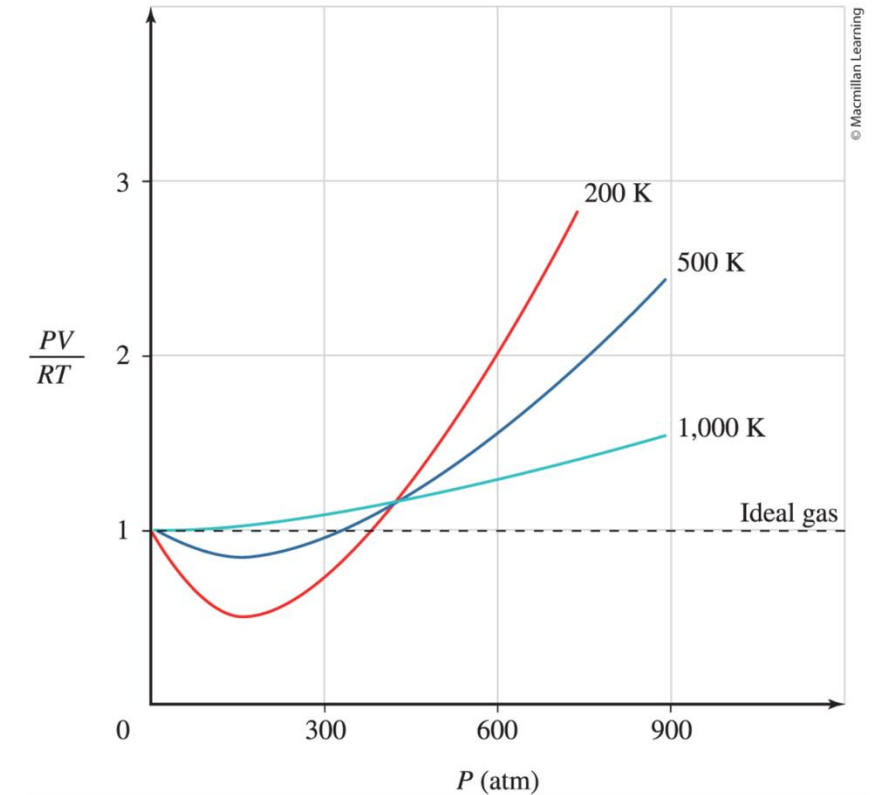


Figure 7.26. Compressibility factor PV/RT as a function of pressure and temperature for N_2 gas.

- Particles of a real gas are attracted to one another, although these attractions tend to be relatively weak because of the long distances between the particles and their rapid speed.
 - At higher temperatures...
 - At lower temperatures...

- The **van der Waals** equation of state is a modification of the ideal gas law that incorporates parameters related to interparticle forces (a) and the volume taken up by gas particles (b).

$$\left(P + a \frac{n^2}{V^2} \right) (V - nb) = nRT$$

- Some of the amazing things we can now do with gases...
 1. Define and interpret pressure; express pressure in a variety of units.
 2. Carry out calculations based on the ideal gas equation of state and the empirical gas laws.
 3. Carry out stoichiometric calculations for gaseous reactions.
 4. Articulate and apply the tenets of the kinetic theory of gases.
 5. Use the kinetic theory to explain the behavior of gases.
 6. Describe the distribution of particle speeds in an ideal gas at equilibrium; predict how changes in molar mass and/or temperature will affect that distribution.
 7. Relate rates of diffusion and effusion to root mean square speed, temperature, and molar mass.
 8. Explain how real gases violate the tenets of the kinetic theory; recognize conditions under which gases behave less ideally.
 9. Apply the van der Waals equation of state to calculate variables related to real gases.